Object Oriented Programming - Previous Year Questions

D Y Patil University, B.Tech Computer Engineering, Sem III - ranked by how often each topic repeats

Papers analysed (4): P1 = Jun 2025 (05/06/2025) | P2 = Nov 2025 (24/11/2025) | P3 = Nov 2025 (19/11/2025, 100 marks) | P4 = Nov 2024 (26/11/2024).
Similar questions are grouped into one topic. **Papers** = in how many of the 4 papers the topic appeared; **Times** = total number of questions on it.

Part 1: Ranking Summary

Rank	Topic	Papers (of 4)	Times	Where	Priority
1	Exception Handling (divide-by-zero programs, try/catch/finally, multiple catch)	4	7	P1, P2, P3, P4	MUST DO
1	OOP Concepts (OOP paradigm, OOP vs structural, Java vs C++, class & object)	4	7	P1, P2, P3, P4	MUST DO
3	Threads (life cycle, multithreading, thread programs)	4	6	P1, P2, P3, P4	MUST DO
4	Types of Inheritance (incl. multiple inheritance)	4	4	P1, P2, P3, P4	MUST DO
4	String / StringBuffer classes and methods	4	4	P1, P2, P3, P4	MUST DO
6	AWT vs Swing, Swing layouts, hierarchy, calculator GUI	3	7	P2, P3, P4	HIGH
7	Method Overloading vs Overriding	3	3	P1, P2, P3	HIGH
7	Abstract class / Interface / Shape program	3	3	P2, P3, P4	HIGH
7	Graphics program (frame / smiley face)	3	3	P1, P3, P4	HIGH
7	Recursion programs (factorial / Fibonacci)	3	3	P1, P3, P4	HIGH
7	JVM working / Java platform independence	3	3	P2, P3, P4	HIGH
7	Array programs (sum/average, matrix)	3	3	P1, P3, P4	HIGH
13	Constructor Overloading (theory + program)	2	2	P1, P3	MEDIUM
13	Multilevel inheritance - volume of sphere	2	2	P1, P3	MEDIUM
13	Access Specifiers	2	2	P3, P4	MEDIUM
13	Argument Passing Mechanism	2	2	P2, P3	MEDIUM
17	Single appearance: Rectangle class	1	1	P1	LOW
17	Single appearance: Sum of first n natural numbers	1	1	P1	LOW
17	Single appearance: nCr and nPr	1	1	P2	LOW
17	Single appearance: Static members and methods	1	1	P2	LOW
17	Single appearance: Packages and types	1	1	P4	LOW
17	Single appearance: Type casting / conversion	1	1	P3	LOW
17	Single appearance: String palindrome	1	1	P3	LOW

Tied topics share the same rank.

Part 2: Topic-wise Questions (in rank order)

#1. Exception Handling (divide-by-zero programs, try/catch/finally, multiple catch) [MUST DO - 4 of 4 papers, 7 times]

Tip: Divide-by-zero program has appeared in ALL 4 papers (10 marks each time). Know it cold.

- **P1 Q3c** (10M): Write a Java program to divide two numbers accepted from user. Handle the number format exception and divide by zero exception using try-catch.
- **P2 Q2e** (5M): Significance of the finally block in exception handling? Can finally exist without try? Explain with example.
- **P2 Q3b** (10M): Write a Java program demonstrating exception handling using try, catch and finally; include ArithmeticException (division by zero).
- **P3 Q1d** (4M): Write a Java program for multiple catch statements.
- **P3 Q5c** (10M): Write a Java program to accept two integers, perform division and demonstrate exception handling using try-catch when divided by zero.
- **P4 Q1e** (5M): With suitable example, explain exception handling in Java.
- **P4 Q3a** (10M): Accept 2 integers, perform division and handle exception using try-catch when divided by zero.

#1. OOP Concepts (OOP paradigm, OOP vs structural, Java vs C++, class & object) [MUST DO - 4 of 4 papers, 7 times]

Tip: Theory opener - nearly every paper has 1-2 of these.

- **P1 Q1a** (5M): Describe features of object oriented programming methodology.
- **P1 Q2a** (5M): Compare and contrast structural programming and object oriented programming.
- **P2 Q2a** (5M): Compare and contrast OOP in Java and C++. Major differences in terms of OOP features?
- **P2 Q2d** (5M): Explain the concept of 'class' and 'object' in OOP with suitable example.
- **P3 Q1a** (4M): Compare and contrast structural programming and object-oriented programming.
- **P3 Q2a** (5M): Explain object-oriented programming paradigm.
- **P4 Q2a** (5M): Explain object-oriented programming paradigm.

#3. Threads (life cycle, multithreading, thread programs) [MUST DO - 4 of 4 papers, 6 times]

Tip: Life cycle appears in P1 & P3; odd/even or 1-10 thread program is the usual coding question.

- **P1 Q1e** (5M): Illustrate life cycle of thread in Java.
- **P1 Q2e** (5M): Write a Java program to demonstrate the making of a thread to print numbers from 1 to 10.
- **P2 Q2b** (5M): Explain the concept of multithreading in Java.
- **P3 Q2e** (5M): Illustrate the life cycle of Thread in Java.
- **P3 Q4c** (10M): Write a Java program with two threads: one displays odd numbers 11 to 20, the other even numbers 1 to 10.
- **P4 Q2e** (5M): What is a thread in Java? How is multithreading implemented in a program?

#4. Types of Inheritance (incl. multiple inheritance) [MUST DO - 4 of 4 papers, 4 times]

Tip: Prepare all types with diagrams + examples (interfaces for multiple).

- **P1 Q2d** (5M): Illustrate types of inheritance in Java.
- **P2 Q1d** (5M): Explain multiple inheritance with suitable example.
- **P3 Q1c** (4M): Illustrate types of Inheritance in Java.
- **P4 Q2d** (5M): Discuss importance of inheritance. List all types of inheritance and explain any two in detail.

#4. String / StringBuffer classes and methods [MUST DO - 4 of 4 papers, 4 times]

Tip: Learn ~5 methods of each with examples.

- **P1 Q2c** (5M): Describe any five methods of StringBuffer class.
- **P2 Q2f** (5M): Explain String and StringBuffer classes with an example.
- **P3 Q2c** (5M): Elaborate the concept of String class with its any 4 methods.
- **P4 Q1c** (5M): Elaborate concept of String class with its any 4 methods.

#6. AWT vs Swing, Swing layouts, hierarchy, calculator GUI [HIGH - 3 of 4 papers, 7 times]

Tip: AWT vs Swing in 3 of 4 papers; layouts in 2; Swing calculator once (10 marks).

- **P2 Q2c** (5M): Explain the difference between AWT and Swing. Why is Swing preferred over AWT?
- **P2 Q3c** (10M): Write a Java Swing program to create a simple calculator GUI (digits 0-9, + - * /, text field for result).
- **P3 Q1e** (4M): Discuss different layouts of Swing package of Java.
- **P3 Q2f** (5M): Differentiate between AWT and Swing package of Java.
- **P3 Q3c** (5M): Write a short note on Java Swing class Hierarchy.
- **P4 Q1f** (5M): Differentiate between AWT and Swing.
- **P4 Q2f** (5M): List and explain various layouts in Java GUI.

#7. Method Overloading vs Overriding [HIGH - 3 of 4 papers, 3 times]

- **P1 Q1d** (5M): Differentiate Method Overloading and Method Overriding.
- **P2 Q3a** (10M): Explain how method overloading differs from method overriding. Write a Java program for overloaded calculateArea() methods (circle, rectangle, triangle).
- **P3 Q2d** (5M): With suitable example, explain method overriding, its usage and rules.

#7. Abstract class / Interface / Shape program [HIGH - 3 of 4 papers, 3 times]

Tip: Abstract vs interface theory + a program using abstract class/interface.

- **P2 Q1b** (5M): Differentiate between abstract class and interface.
- **P3 Q5b** (10M): Write a Java program to create abstract class Shape. Derive sphere, cone and cylinder. Calculate area and volume (method overriding).
- **P4 Q3b** (10M): University grace-marks application: keep academic marks and sports grace marks separate, generate total; invoke methods of base class & interface using reference.

#7. Graphics program (frame / smiley face) [HIGH - 3 of 4 papers, 3 times]

Tip: 'All the best' in a Frame appeared twice.

- **P1 Q1f** (5M): Write a graphics Java program to display "All the best" using frame.
- **P3 Q3f** (5M): Write a graphics Java program to display "All the Best" using frame.
- **P4 Q3c** (10M): Write a Java program to draw a smiling face using Graphics class methods.

#7. Recursion programs (factorial / Fibonacci) [HIGH - 3 of 4 papers, 3 times]

Tip: Factorial 2x, Fibonacci 1x - practise both.

- **P1 Q2b** (5M): Write a program to print Factorial of a number using recursion.
- **P3 Q2b** (5M): Write a Java program to print factorial of a number.
- **P4 Q1b** (5M): Write a program to print Fibonacci series using recursion.

#7. JVM working / Java platform independence [HIGH - 3 of 4 papers, 3 times]

- **P2 Q1a** (5M): Explain how Java is platform independent.
- **P3 Q3a** (5M): Explain the working of JVM in detail.
- **P4 Q1a** (5M): Explain the detail working of JVM.

#7. Array programs (sum/average, matrix) [HIGH - 3 of 4 papers, 3 times]

Tip: Sum & average of array repeated in P3 and P4.

- **P1 Q1c** (5M): Write a Java program to accept m*n matrix and display it in natural form.
- **P3 Q3b** (5M): Write a Java program to calculate the sum and average of array elements with user input.
- **P4 Q2c** (5M): Write a program to calculate sum and average of array elements in Java with user input.

#13. Constructor Overloading (theory + program) [MEDIUM - 2 of 4 papers, 2 times]

Tip: 10 marks both times.

- **P1 Q3a** (10M): What is constructor overloading? Write a program to demonstrate constructor overloading.
- **P3 Q4a** (10M): What is constructor overloading? Write a Java program to demonstrate constructor overloading.

#13. Multilevel inheritance - volume of sphere [MEDIUM - 2 of 4 papers, 2 times]

Tip: 10 marks both times.

- **P1 Q3b** (10M): Write a Java program to calculate volume of sphere using multilevel inheritance (base class accepts radius; derived class finds area of circle; next derived class calculates and displays volume).
- **P3 Q4b** (10M): Write a Java program to calculate volume of a sphere using multilevel inheritance (radius -> area of sphere -> volume).

#13. Access Specifiers [MEDIUM - 2 of 4 papers, 2 times]

- **P3 Q1b** (4M): Explain the different access specifiers in Java.
- **P4 Q2b** (5M): Explain the different access specifiers in Java.

#13. Argument Passing Mechanism [MEDIUM - 2 of 4 papers, 2 times]

- **P2 Q1c** (5M): Explain Argument Passing Mechanism with suitable example.
- **P3 Q5a** (10M): Elaborate argument passing mechanism. Write a Java program for the different cases along with output.

#17. Single appearance: Rectangle class [LOW - 1 of 4 papers, 1 times]

- **P1 Q1b** (5M): Write a Java program to create class Rectangle and calculate area and perimeter of it.

#17. Single appearance: Sum of first n natural numbers [LOW - 1 of 4 papers, 1 times]

- **P1 Q2f** (5M): Write a Java program to calculate and display the sum of first n natural numbers.

#17. Single appearance: nCr and nPr [LOW - 1 of 4 papers, 1 times]

- **P2 Q1e** (5M): Write a program in Java to find nCr and nPr.

#17. Single appearance: Static members and methods [LOW - 1 of 4 papers, 1 times]

- **P2 Q1f** (5M): Explain static data members and methods.

#17. Single appearance: Packages and types [LOW - 1 of 4 papers, 1 times]

- **P4 Q1d** (5M): What is package? Explain types of packages.

#17. Single appearance: Type casting / conversion [LOW - 1 of 4 papers, 1 times]

- **P3 Q3d** (5M): Describe the concept of type casting and type conversion in Java.

#17. Single appearance: String palindrome [LOW - 1 of 4 papers, 1 times]

- **P3 Q3e** (5M): Write a program in Java to check whether a string is palindrome or not.

Part 3: All Questions, Paper-wise

P1 - Jun 2025 (05/06/2025)

Q	Question	Marks
Q1a	Describe features of object oriented programming methodology.	5
Q1b	Write a Java program to create class Rectangle and calculate area and perimeter of it.	5
Q1c	Write a Java program to accept m*n matrix and display it in natural form.	5
Q1d	Differentiate Method Overloading and Method Overriding.	5
Q1e	Illustrate life cycle of thread in Java.	5
Q1f	Write a graphics Java program to display "All the best" using frame.	5
Q2a	Compare and contrast structural programming and object oriented programming.	5
Q2b	Write a program to print Factorial of a number using recursion.	5
Q2c	Describe any five methods of StringBuffer class.	5
Q2d	Illustrate types of inheritance in Java.	5
Q2e	Write a Java program to demonstrate the making of a thread to print numbers from 1 to 10.	5
Q2f	Write a Java program to calculate and display the sum of first n natural numbers.	5
Q3a	What is constructor overloading? Write a program to demonstrate constructor overloading.	10
Q3b	Write a Java program to calculate volume of sphere using multilevel inheritance (base class accepts radius; derived class finds area of circle; next derived class calculates and displays volume).	10
Q3c	Write a Java program to divide two numbers accepted from user. Handle the number format exception and divide by zero exception using try-catch.	10

P2 - Nov 2025 (24/11/2025)

Q	Question	Marks
Q1a	Explain how Java is platform independent.	5
Q1b	Differentiate between abstract class and interface.	5
Q1c	Explain Argument Passing Mechanism with suitable example.	5
Q1d	Explain multiple inheritance with suitable example.	5
Q1e	Write a program in Java to find nCr and nPr.	5
Q1f	Explain static data members and methods.	5
Q2a	Compare and contrast OOP in Java and C++. Major differences in terms of OOP features?	5
Q2b	Explain the concept of multithreading in Java.	5
Q2c	Explain the difference between AWT and Swing. Why is Swing preferred over AWT?	5
Q2d	Explain the concept of 'class' and 'object' in OOP with suitable example.	5
Q2e	Significance of the finally block in exception handling? Can finally exist without try? Explain with example.	5
Q2f	Explain String and StringBuffer classes with an example.	5
Q3a	Explain how method overloading differs from method overriding. Write a Java program for overloaded calculateArea() methods (circle, rectangle, triangle).	10
Q3b	Write a Java program demonstrating exception handling using try, catch and finally; include ArithmeticException (division by zero).	10
Q3c	Write a Java Swing program to create a simple calculator GUI (digits 0-9, + - * /, text field for result).	10

P3 - Nov 2025 (19/11/2025, 100 marks)

Q	Question	Marks
Q1a	Compare and contrast structural programming and object-oriented programming.	4
Q1b	Explain the different access specifiers in Java.	4

Q	Question	Marks
Q1c	Illustrate types of Inheritance in Java.	4
Q1d	Write a Java program for multiple catch statements.	4
Q1e	Discuss different layouts of Swing package of Java.	4
Q2a	Explain object-oriented programming paradigm.	5
Q2b	Write a Java program to print factorial of a number.	5
Q2c	Elaborate the concept of String class with its any 4 methods.	5
Q2d	With suitable example, explain method overriding, its usage and rules.	5
Q2e	Illustrate the life cycle of Thread in Java.	5
Q2f	Differentiate between AWT and Swing package of Java.	5
Q3a	Explain the working of JVM in detail.	5
Q3b	Write a Java program to calculate the sum and average of array elements with user input.	5
Q3c	Write a short note on Java Swing class Hierarchy.	5
Q3d	Describe the concept of type casting and type conversion in Java.	5
Q3e	Write a program in Java to check whether a string is palindrome or not.	5
Q3f	Write a graphics Java program to display "All the Best" using frame.	5
Q4a	What is constructor overloading? Write a Java program to demonstrate constructor overloading.	10
Q4b	Write a Java program to calculate volume of a sphere using multilevel inheritance (radius -> area of sphere -> volume).	10
Q4c	Write a Java program with two threads: one displays odd numbers 11 to 20, the other even numbers 1 to 10.	10
Q5a	Elaborate argument passing mechanism. Write a Java program for the different cases along with output.	10
Q5b	Write a Java program to create abstract class Shape. Derive sphere, cone and cylinder. Calculate area and volume (method overriding).	10
Q5c	Write a Java program to accept two integers, perform division and demonstrate exception handling using try-catch when divided by zero.	10

P4 - Nov 2024 (26/11/2024)

Q	Question	Marks
Q1a	Explain the detail working of JVM.	5
Q1b	Write a program to print Fibonacci series using recursion.	5
Q1c	Elaborate concept of String class with its any 4 methods.	5
Q1d	What is package? Explain types of packages.	5
Q1e	With suitable example, explain exception handling in Java.	5
Q1f	Differentiate between AWT and Swing.	5
Q2a	Explain object-oriented programming paradigm.	5
Q2b	Explain the different access specifiers in Java.	5
Q2c	Write a program to calculate sum and average of array elements in Java with user input.	5
Q2d	Discuss importance of inheritance. List all types of inheritance and explain any two in detail.	5
Q2e	What is a thread in Java? How is multithreading implemented in a program?	5
Q2f	List and explain various layouts in Java GUI.	5
Q3a	Accept 2 integers, perform division and handle exception using try-catch when divided by zero.	10
Q3b	University grace-marks application: keep academic marks and sports grace marks separate, generate total; invoke methods of base class & interface using reference.	10
Q3c	Write a Java program to draw a smiling face using Graphics class methods.	10